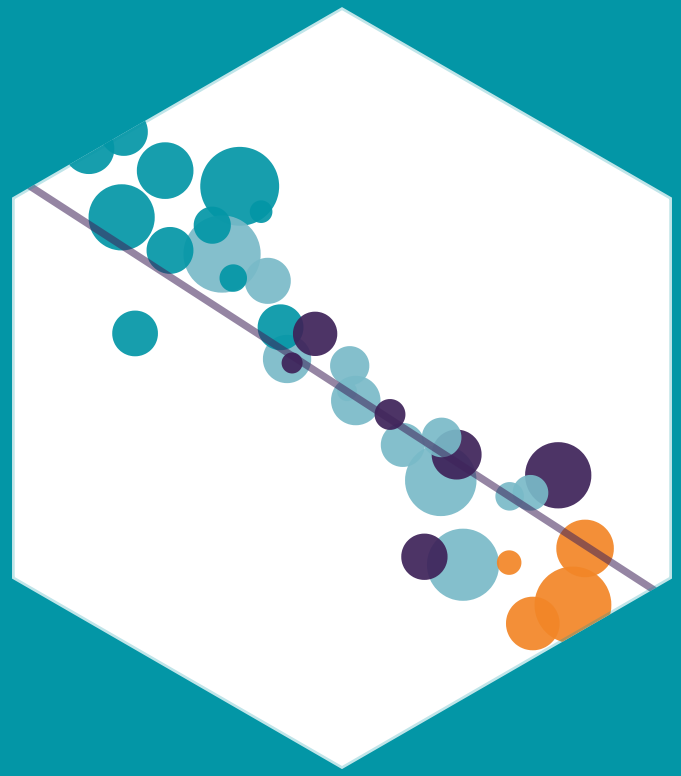
ECONOMETRICS

TA SESSION 6 · PART I — RCTS

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Planification — 2° Part

TA6 — Part I: RCTs ← today · Part II: Matching (Thursday)

📅 Hand in Assignment 6 by **17th Nov**

TA7 — IV, RDD

📅 Hand in Assignment 7 by **25th Nov**

TA8 — Difference-in-Differences

📅 Hand in Assignment 8 by **2nd Dec**

TA9 — Review for the Exam

🗓️ Week of **Dec 8**

Today

1. **Potential outcomes** — the fundamental problem of causality *(birthweight)*
2. **Randomization** — why it solves the problem
3. **Ham & LaLonde (1996)** — a real RCT, in Python
4. **Designing an A/B test** — exam questions, one by one *(a ride-hailing app)*

📓 Everything in **TA6_RCT.ipynb** · data: **TA6.dta**, **app_experiment.csv**

Potential outcomes

- $D_i \in \{0, 1\}$: treatment indicator
- $Y_i(1)$: outcome of i **with** treatment
- $Y_i(0)$: outcome of i **without** treatment
- Individual causal effect: $\tau_i = Y_i(1) - Y_i(0)$

What we observe:

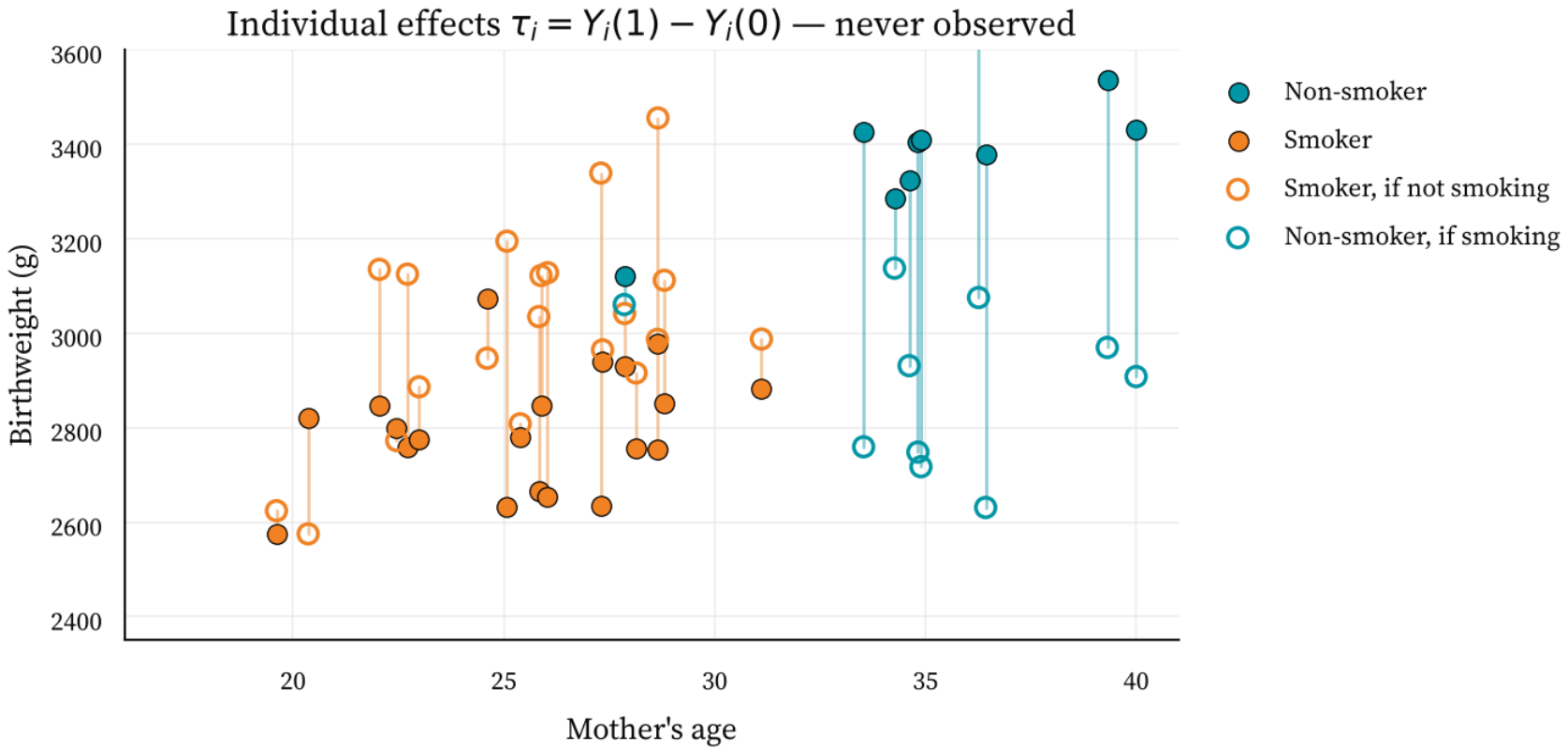
$$Y_i = D_i\, Y_i(1) + (1 - D_i)\, Y_i(0)$$

The fundamental problem of causal inference

We **never** observe both $Y_i(1)$ and $Y_i(0)$ for the same person.

Causality = a **missing data** problem.

Example: does smoking during pregnancy reduce birthweight?



Each vertical segment is one τ_i. We only ever see **one end** of it.

Two averages of τ_i: ATE and ATT

$$\text{ATE} = E\big[Y_i(1) - Y_i(0)\big] \qquad \text{ATT} = E\big[Y_i(1) - Y_i(0) \,\big|\, D_i = 1\big]$$

ATE: average over **everyone**.

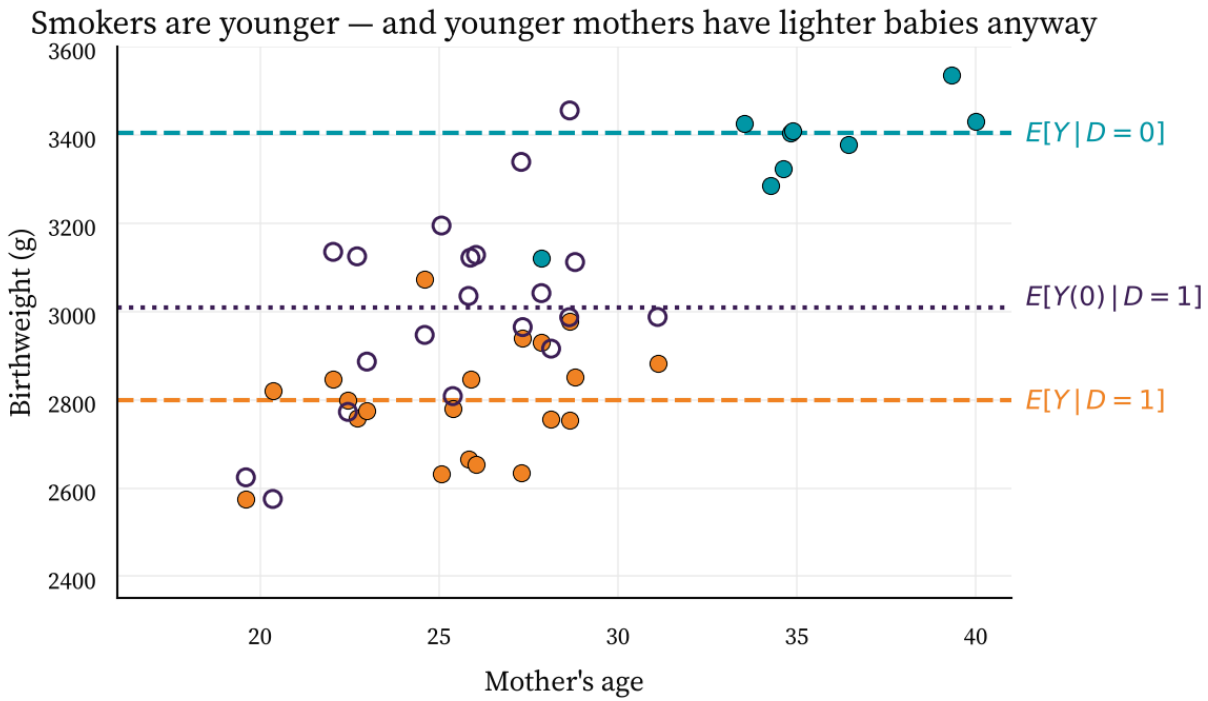
"What if all mothers smoked vs none?"

ATT: average over **those treated** only.

"How much lighter are smokers' babies because their mothers smoked?"

In our 30 mothers: ATE = **-310 g**, ATT = **-210 g**. They differ because (i) the effect varies with age and (ii) smokers are younger than the average mother. Same question, **different population**.

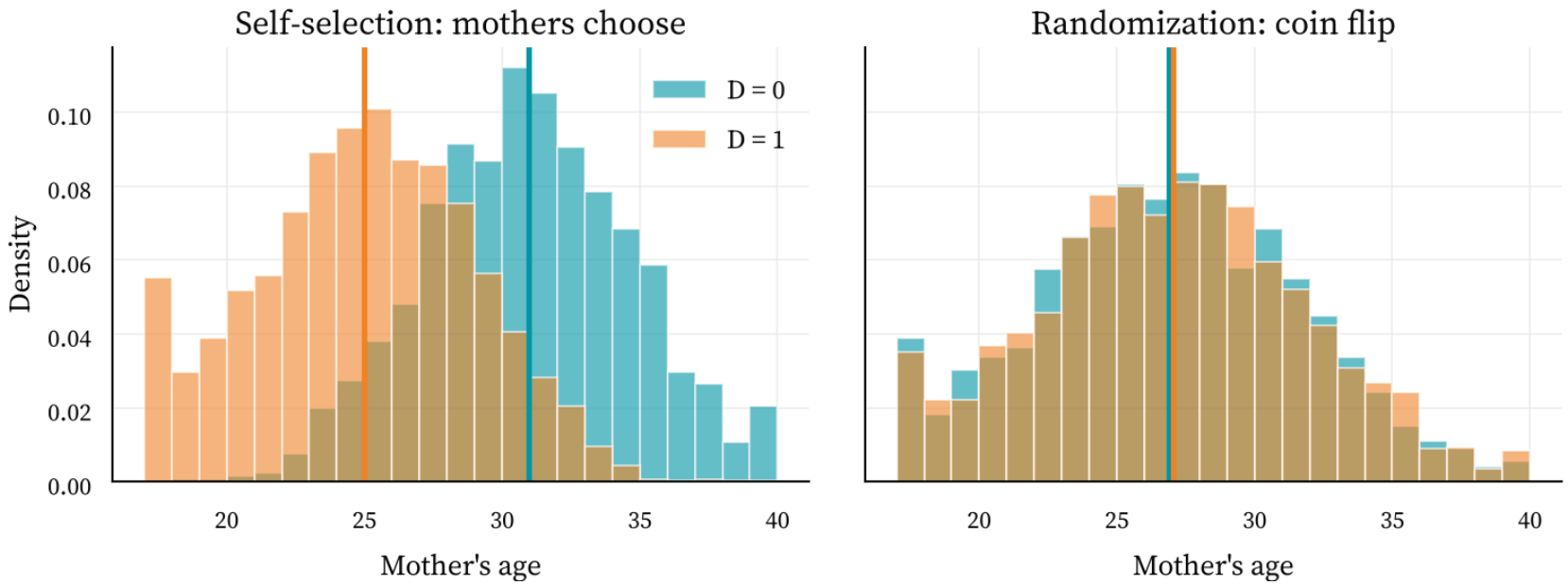
The naive comparison is biased



$$\underbrace{E[Y|D=1] - E[Y|D=0]}_{\text{naive}=-606} = \underbrace{E[Y(1) - Y(0)|D=1]}_{\text{ATT}=-210} + \underbrace{E[Y(0)|D=1] - E[Y(0)|D=0]}_{\text{selection bias}=-396}$$

Randomization kills selection bias

Randomly assign D_i (coin flip) $\Rightarrow$ $\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp D_i$



Same distribution of **everything** (observed or not) in both groups ⇒ $E[Y(0) \mid D = 1] = E[Y(0) \mid D = 0]$ ⇒ selection bias $= 0$, and $\text{ATE} = \text{ATT} = E[Y \mid D = 1] - E[Y \mid D = 0]$.

Does job training raise earnings?

- **Outcome:** `re78` — real earnings in 1978 ($1000s) · **Treatment:** `train` — randomly assigned
- **Pre-treatment covariates:** age, education, race, marital status, earnings in 1974–75

train	age	educ	black	hisp	married	nodegree	re74	re75	re78
1	37	11	1	0	1	1	0.00	0.00	9.93
1	22	9	0	1	0	1	0.00	0.00	3.60
1	30	12	1	0	0	0	0.00	0.00	24.91
1	27	11	1	0	0	1	0.00	0.00	7.51
1	33	8	1	0	0	1	0.00	0.00	0.29

```
df = pd.read_stata("TA6.dta")
df.groupby("train")["re78"].agg(["mean", "std", "count"])
```

Difference in means = regression on D

Mean, treated (185)	6.35
Mean, control (260)	4.55
Difference	**1.79**
t (unequal var.)	2.67 ($p = 0.008$)

$$\texttt{earnings}_i = \beta_0 + \beta_1 D_i + \varepsilon_i$$

	coef.	s.e.
`treat`	**1.794*****	(0.633)
`Intercept`	4.555	(0.408)

```
stats.ttest_ind(df.re78[df.train == 1], df.re78[df.train == 0], equal_var=False)
smf.ols("re78 ~ train", data=df).fit().summary()
```

$\hat{\beta}_0$ = control mean, $\hat{\beta}_1$ = difference in means: training raises 1978 earnings by ≈ **$1,800**.

Balance: are treated and controls comparable?

	Treated	Control	Diff.	p
`re74`	2.10	2.11	-0.011	0.982
`re75`	1.53	1.27	+0.265	0.385
`age`	25.82	25.05	+0.762	0.266
`educ`	10.35	10.09	+0.257	0.150
`nodegree`	0.71	0.83	**-0.127*****	0.002
`married`	0.19	0.15	+0.035	0.334
`black`	0.84	0.83	+0.016	0.647
`hisp`	0.06	0.11	**-0.048***	0.064

- Treated are **less likely to lack a high-school degree**; slightly more Hispanic (10%).
- With 8 tests, ~1 "significant" difference is expected **by chance**.
- Joint test (`treat` on all X): $F = 2.14$, $p = 0.031$ → mild imbalance ⇒ **check robustness with controls** (next slide).

Adding covariates

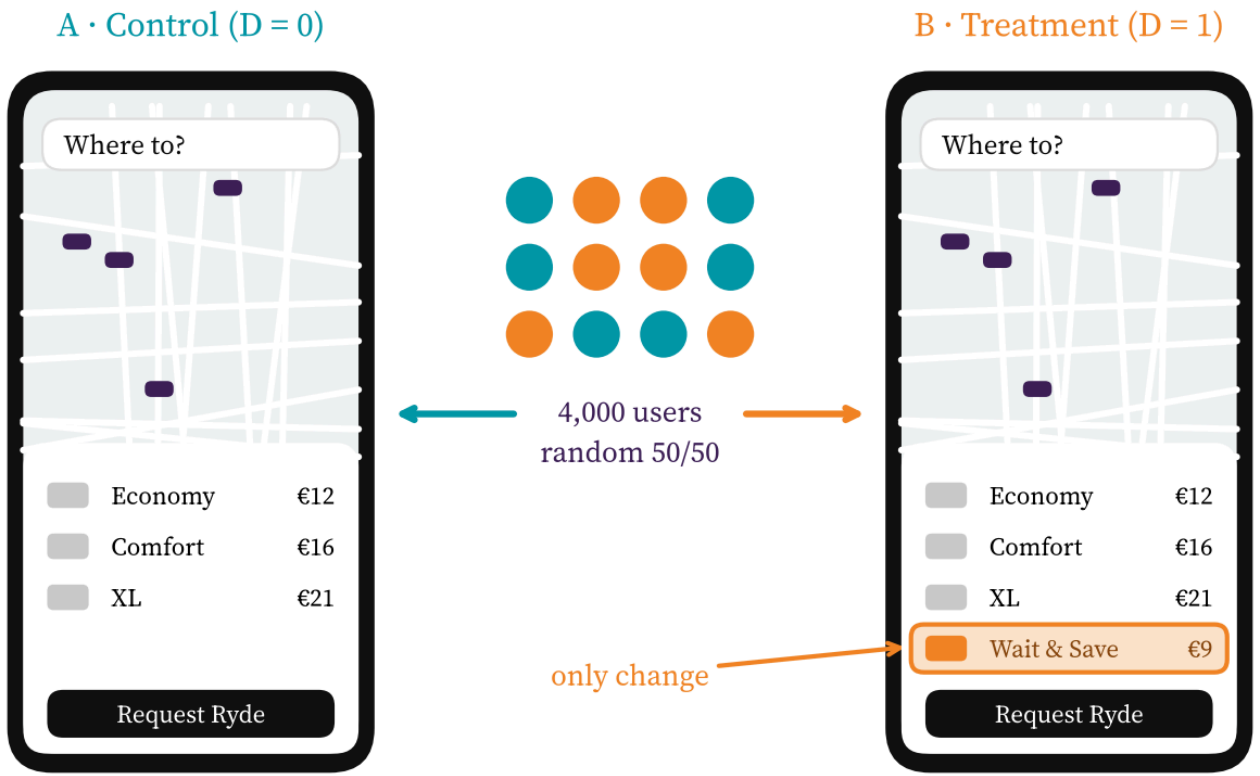
	(1) Simple	(2) + Controls
`treat`	1.794***	1.625**
	(0.633)	(0.640)
Controls	No	Yes
R^2	0.018	0.049

- Estimate barely moves (1.79 → 1.63): consistent with successful randomization.
- **Regression adjustment** (one regression per group, predict $\hat{Y}_i(1)$ and $\hat{Y}_i(0)$ for everyone, average the difference): ATE = **1.54** (all men) · ATT = **1.76** (trained men only).

```
smf.ols("re78 ~ train + re74 + re75 + age + agesq + nodegree + married + black + hisp", data=df).fit()
```

The case: one new option in the app

Ryde (a fictional ride-hailing app) wants to add a cheaper option: **"Wait & Save"**. Will it increase rides?



Q1 · How would you implement the experiment?

Exam

Describe briefly how you would implement the experiment. At which level should you randomize (country, city, user)? Are you worried about take-up, and why?

- **Units:** active users. **Treatment** $D_i = 1$ if user i sees app B (with "Wait & Save"), assigned at random 50/50.
- **Outcome** Y_i: rides in the 4 weeks of the test. **Covariates** X_i (measured before): age, gender, city, rides in the previous 4 weeks.
- **Parameter:** $\text{ATE} = E[Y_i(1) - Y_i(0)]$, estimated by $\bar{Y}_B - \bar{Y}_A$.
- **Level:** *country* → too few, too different units. *User* → many units, high power. But users in the same city share drivers: if B users take more rides, A users wait longer (**SUTVA**) → if that matters, randomize by **city** and cluster SEs.
- **Take-up:** yes — we can *offer* the option, not force its use. Users who pick it are self-selected (heavy users) → compare by **assignment** (ITT), never choosers vs non-choosers.

Q2 · Formulas for the ATE and ATT; how to estimate them?

Exam

Give the formulas for the ATE and ATT. How would you estimate them?

$$\text{ATE} = E[Y_i(1) - Y_i(0)] \qquad \text{ATT} = E[Y_i(1) - Y_i(0) \mid D_i = 1]$$

Randomization ⇒ $D_i \perp (Y_i(1), Y_i(0))$ ⇒ $E[Y(0) \mid D = 1] = E[Y(0) \mid D = 0]$, so

$$\widehat{\text{ATE}} = \widehat{\text{ATT}} = \bar{Y}_{D=1} - \bar{Y}_{D=0} = \hat{\beta}_1 \text{ in } Y_i = \beta_0 + \beta_1 D_i + \varepsilon_i$$

In our data: $\hat{\beta}_1$ = **0.33** rides (robust s.e. 0.14). (Simulated truth: ATE = 0.40, ATT = 0.41 — equal up to sampling noise.)

```
smf.ols("rides_post ~ treat", data=app).fit(cov_type="HC1")
```

Q3 · Are treatment and control truly random?

Exam

How could you check if treatment and control groups are truly random?

	B	A	p-value
Age	34.25	34.44	0.49
Female	0.53	0.54	0.48
Rides before	3.97	3.98	0.98
Madrid	0.28	0.29	0.50
Barcelona	0.32	0.31	0.52
Paris	0.20	0.20	0.84
London	0.20	0.20	0.85

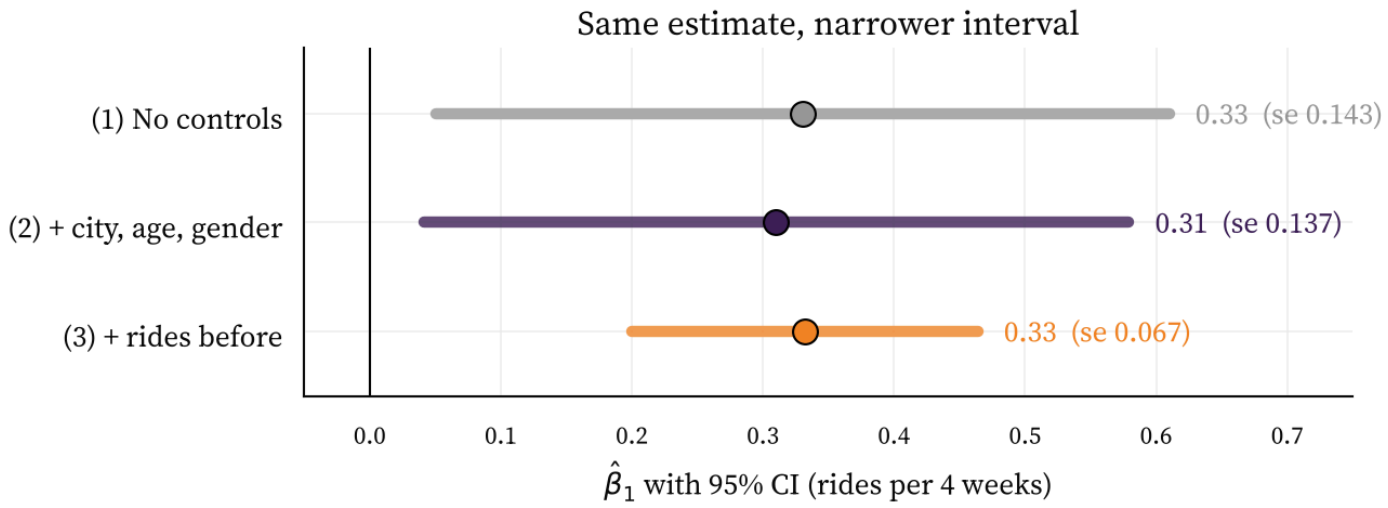
- Compare **pre-treatment** variables (incl. the pre-period outcome) with t-tests.
- **Joint test:** regress D on all X: $F = 0.27$, $p = 0.95$ → X do not predict D ✅
- A few rejections are expected by chance (1 in 20 at 5%).

Q4 · Would you use age, gender, city?

Exam

Explain briefly how you would calculate the ATE. Suppose you had information on age, gender, city: would you use it?

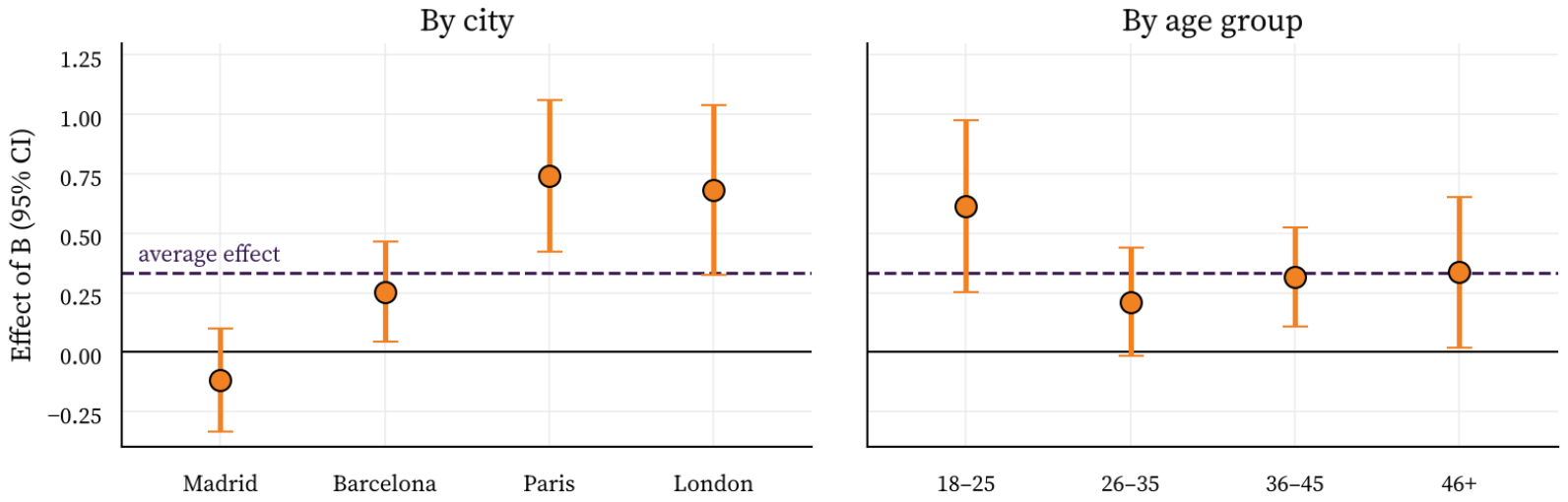
$$\texttt{rides}_i = \beta_0 + \beta_1 D_i + X_i'\gamma + \varepsilon_i$$



Yes, but not for unbiasedness: $D \perp X$ by design → the **estimate does not change**. They reduce residual variance → **smaller SE** (more power). And they let us estimate **heterogeneous effects** (next slide). Only **pre-treatment** variables!

Q4 · Heterogeneous treatment effects

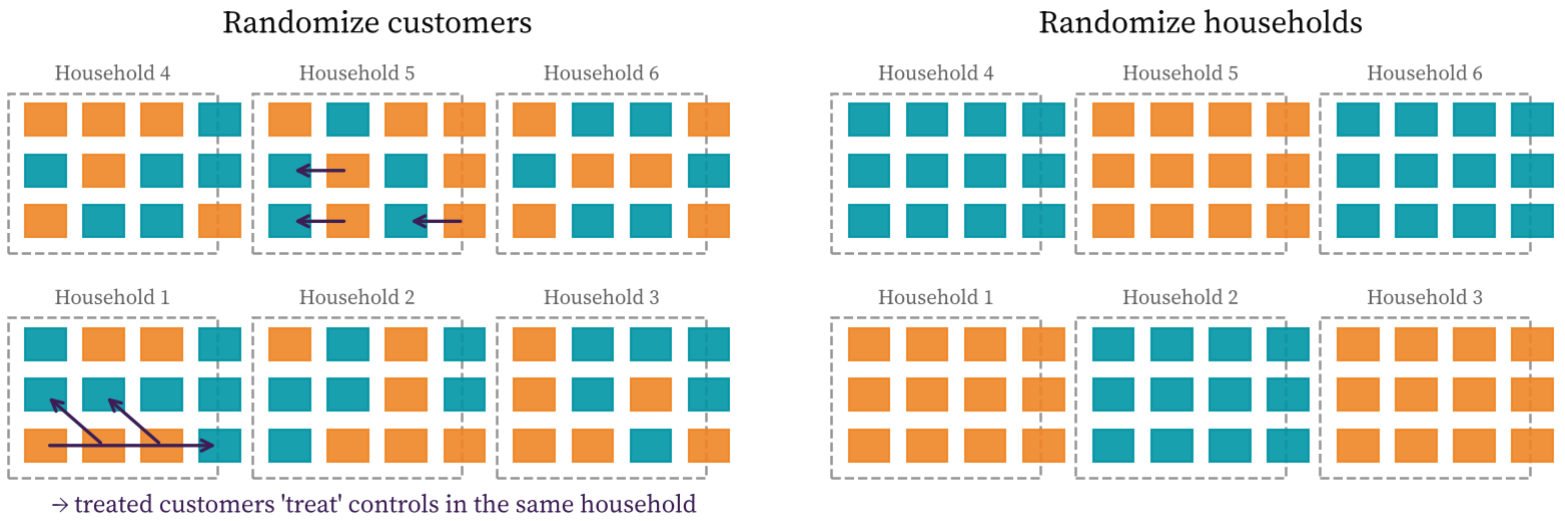
$$\texttt{rides}_i = \beta_0 + \beta_1 D_i + \sum_c \delta_c \left(D_i \times \text{city}_c\right) + X_i'\gamma + \varepsilon_i$$



The option helps most in **London/Paris** and among **young users** → target the launch. Randomization holds within each subgroup, so each subgroup effect is causal (but pre-specify subgroups: slice 20 ways and one will look significant by chance).

Q5 · When SUTVA fails, change the unit of randomization

SUTVA: my outcome depends only on **my own** treatment, not on others'.



Fix: randomize at the level where the interaction happens (**household, city, village**) so spillovers stay *within* one arm; measure the outcome at that level; **cluster SEs** at that level. *Ryde:* users share drivers within a city → randomize by city.

Practice (5 min, in pairs)

Exam-style question

An internet provider **randomly contacts half** of its phone customers with an offer for a high-speed internet package.

a. How would you calculate the ATE? Would you use age, gender, race?

b. *(Bonus)* Some customers live in the **same household** and share an internet connection. Is that an issue? How would you address it in the randomization?

(a) D = contacted, Y = bought internet: $\widehat{\text{ATE}} = \bar{Y}_{D=1} - \bar{Y}_{D=0}$ = OLS of Y on D. Covariates: same estimate, smaller SE, heterogeneity (careful: targeting by race is legally/ethically problematic).

(b) Yes, SUTVA fails: a contacted customer who buys gives internet to non-contacted members of the household → control outcomes affected → bias toward zero. **Randomize by household** (all members get the same assignment), **cluster SEs by household**.

Summary

- We never see $Y_i(1)$ and $Y_i(0)$ together $\rightarrow$ naive comparisons mix the effect with **selection bias**.
- **Randomization** $\Rightarrow$ selection bias $= 0 \Rightarrow$ difference in means = ATE = ATT.
- **Check balance** on pre-treatment variables (individual + joint test).
- **Covariates:** same estimate, **smaller SE**, and **heterogeneous effects**.
- Interactions between units (**SUTVA**) $\rightarrow$ randomize at the cluster level.
- Imperfect take-up $\rightarrow$ compare by **assignment** (ITT).

Thursday: what if we *cannot* randomize? $\rightarrow$ Matching